

flspcell_fast_1p65V_m40C Library

Cell Groups
DFRBPQ

DFRBPQ



*flspcell_fast_1p65V_m40C Cell Library: Process
flspcell_fast_1p65V_m40C, Voltage 1.65, Temp -40.00*

Truth Table

INPUT			OUTPUT
CLK	D	RESET_B	Q
R	0	1	0
R	1	1	1
x	x	0	0
x	x	1	IQ

Footprint

Cell Name	Area
sg13g2_dfrbpq1arx_1	48.98880

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	CLK	D	RESET_B	Q
sg13g2_dfrbpq1arx_1	0.02289	0.01562	0.02869	0.30000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_dfrbpq1arx_1	20547.30000	24051.80000	26526.50000

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK->Q (RR)	0.01860	0.00100	0.17661	0.32940	0.06480	0.31119	2.50740	0.30000	0.71440

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK->Q (RF)	0.01860	0.00100	0.19125	0.32940	0.06480	0.32421	2.50740	0.30000	0.66115
	RESET_B->Q (FF)	0.01860	0.00100	0.22410	0.32940	0.06480	0.38670	2.50740	0.30000	0.85286

Constraint Information

Constraints(ns) for CLK rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	CLK ()	0.01860	0.00000	0.05417	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Constraints(ns) for CLK falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	CLK ()	0.01860	0.00000	0.05737	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	hold	CLK (R)	0.01860	0.01860	-0.02690	1.26300	1.26300	-0.10794	2.50740	2.50740	-0.15053
	setup	CLK (R)	0.01860	0.01860	0.22251	1.26300	1.26300	0.24285	2.50740	2.50740	0.22727

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	hold	CLK (R)	0.01860	0.01860	-0.01467	1.26300	1.26300	-0.10794	2.50740	2.50740	-0.17414
	setup	CLK (R)	0.01860	0.01860	0.23474	1.26300	1.26300	0.32110	2.50740	2.50740	0.39255

Constraints(ns) for RESET_B rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	recovery	CLK (R)	0.01860	0.01860	0.05379	1.26300	1.26300	0.18079	2.50740	2.50740	0.29220
	removal	CLK (R)	0.01860	0.01860	-0.04890	1.26300	1.26300	-0.17539	2.50740	2.50740	-0.28925

Constraints(ns) for RESET_B falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	RESET_B ()	0.01860	0.00000	0.05737	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.00100	0.28720	0.32940	0.06480	0.32842	2.50740	0.30000	0.72557

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.00100	0.31393	0.32940	0.06480	0.36033	2.50740	0.30000	0.73335
	RESET_B	0.01860	0.00100	0.29495	0.32940	0.06480	0.31687	2.50740	0.30000	0.45631

Passive power(pJ) for CLK rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.05848	0.32940	0.09886	2.50740	0.46605

Passive power(pJ) for CLK falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.12059	0.32940	0.16336	2.50740	0.53668

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(D * RESET_B * Q)	0.01860	0.03920	0.32940	0.07976	2.50740	0.44762
	(D * !RESET_B * * !Q)	0.01860	0.05848	0.32940	0.09886	2.50740	0.46605
	(!D * RESET_B * !Q)	0.01860	0.06096	0.32940	0.10154	2.50740	0.46915
	(!D * !RESET_B * * !Q)	0.01860	0.07130	0.32940	0.11170	2.50740	0.47897

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(D * RESET_B * Q)	0.01860	0.09760	0.32940	0.14039	2.50740	0.51355
	(D * RESET_B * !Q)	0.01860	0.12059	0.32940	0.16336	2.50740	0.53668
	(D * !RESET_B * * !Q)	0.01860	0.05620	0.32940	0.09759	2.50740	0.45709
	(!D * RESET_B * Q)	0.01860	0.34758	0.32940	0.37984	2.50740	0.73616
	(!D * RESET_B * !Q)	0.01860	0.04925	0.32940	0.09077	2.50740	0.45034
	(!D * !RESET_B * * !Q)	0.01860	0.04362	0.32940	0.08492	2.50740	0.44439

Passive power(pJ) for D rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.10128	0.32940	0.11629	2.50740	0.26668

Passive power(pJ) for D falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.10788	0.32940	0.12378	2.50740	0.27395

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.10128	0.32940	0.11629	2.50740	0.26668
	(!CLK * RESET_B)	0.01860	0.17486	0.32940	0.19069	2.50740	0.35045
	(!CLK * !RESET_B)	0.01860	0.10687	0.32940	0.11687	2.50740	0.22284

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.10788	0.32940	0.12378	2.50740	0.27395
	(!CLK * RESET_B)	0.01860	0.14514	0.32940	0.16165	2.50740	0.32098
	(!CLK * !RESET_B)	0.01860	0.08446	0.32940	0.09512	2.50740	0.20086

Passive power(pJ) for RESET_B rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.00934	0.32940	0.02220	2.50740	0.15639

Passive power(pJ) for RESET_B falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.04744	0.32940	0.06625	2.50740	0.28237

Passive power(pJ) for RESET_B rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(CLK * D * !Q)	0.01860	0.00934	0.32940	0.02220	2.50740	0.15639
	(CLK * !D * !Q)	0.01860	0.00000	0.32940	0.00000	2.50740	0.00000
	(!CLK * D * !Q)	0.01860	0.07812	0.32940	0.09650	2.50740	0.31118
	(!CLK * !D * !Q)	0.01860	0.00486	0.32940	0.00491	2.50740	0.00491

Passive power(pJ) for RESET_B falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(CLK * D * !Q)	0.01860	0.34207	0.32940	0.38343	2.50740	0.74387
	(CLK * !D * !Q)	0.01860	0.00000	0.32940	0.00000	2.50740	0.00000
	(!CLK * D * !Q)	0.01860	0.04744	0.32940	0.06625	2.50740	0.28237
	(!CLK * !D * !Q)	0.01860	-0.00486	0.32940	-0.00491	2.50740	-0.00491